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Project Title:	The Impact of Microgravity on Streptococcus pneumoniae Gene Expression and Virulence Potential		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	GSRP participant moved to co-P.I. lab (Dr. Alfredo Torres) in January 2005 to extend research after project objective 1 was completed. Objective 2 was updated to take advantage of the E. coli genetic systems to study the effects of low-shear modeled microgravity on a clinical E. coli isolate from a Crohn's Disease patient using the ground model system utilized for objective 1. The relevance of this 2nd objective allowed an in-depth focus on understanding the effects of mechanically-based forces (e.g. low-shear stress, reduced gravity), as experienced in space, on the physiology and pathogenic potential of a novel and unique E. coli isolate possessing adherent and invasive characteristics thought to contribute to the development of inflammatory bowel disease.		
COI Name (Institution):	Torres, Alfredo (University of Texas Medical Branch at Galveston)		
Grant/Contract No.:			
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Task Description:	Critical to the success of long-term space flight is the prevention of human disease. Infectious diseases have impacted past missions. There is a documented history of infections during space flight that include those impacting the respiratory tract, skin, urinary tract, and gastrointestinal system (1). While the responsible microbes are carried onboard by the crew, it is not clearly understood how the response / adaptation of these microbes to this unique environment alter their pathogenic potential. Bacteria responsible for such infections have often been associated with normal flora or carrier states, which express new properties or exploit a declining immune system as opportunistic pathogens (2-3). Evidence is accumulating that these bacteria undergo physiological and virulence-associated alterations in microgravity and the space environment which include increases in growth rate and biomass, structural changes in cell wall thickness, and reduced antibiotic sensitivity (4-7). Bacteria exhibit altered physiology and virulence properties in microgravity environments yet the mechanisms responsible for these adaptations remain unknown. While many of the earlier studies have primarily focused on well-characterized enteric species, little is known regarding how respiratory pathogens are affected by the microgravity environment. Streptococcus pneumoniae is a significant respiratory pathogen in individuals with compromised immunity (e.g. young, elderly, shuttle crewmembers) and a major cause of meningitis, bacterial pneumonia, bacteremia, otitis media, and sinusitis. Streptococcus species have been isolated from crewmembers prior to space flight and from air samples taken during flight representing a potential health risk to crewmembers during extended space habitation (5, 8). Little is known about the responses of opportunistic pathogens such as S. pneumoniae to the space environment. In objective 1 of this study, we investigated the effects of low-shear modeled microgravity (LSMMG) on the global gene expression of the opportunistic pathogen Streptococcus pneumoniae using DNA microarray analysis to understand how alterations in the genomic profile of this microbe could influence its virulence potential. Information on bacterial gene expression will advance our understanding of the regulation of bacterial virulence in this unique environment and the underlying pathogenic mechanisms. While most of the bacterial species studied thus far have been well-characterized strains, little is known regarding the effects of LSMMG and the space environment on disease-related clinical isolates. In objective 2 of this study we examined the effects of LSMMG on the physiology and colonization potential of a clinical Escherichia coli isolate obtained from a Crohn's Disease patient which has been shown to exhibit inherent adhesive and invasive properties. These clinical isolates (adherent-invasive E. coli; AIEC) have been suggested to contribute to the etiology of inflammatory bowel diseases and little is understood regarding the effects of mechanically-based forces such as low-shear and reduced gravitational conditions as observed during space flight. The findings from objective 2 will enhance our understanding of how clinical isolates associated with human disease may be altered by space flight leading to increased health risks. The results from objective 1 will help us in identification of novel and unique regulatory mechanisms used in the space environment while objective 2 results will provide important insight into how uncharacterized clinical isolates are affected by LSMMG conditions as experienced in space. This information will be important to establish to generate a rational risk assessment of infection and disease in the space environment. References: 1. Ferguson, J., Taylor, G., Mieszkuc, B. 1975. SP-368 Biomedical Results of Apollo. pp.83-103. 2. Nefedov, Y., Shilov, V., Konstantinova, I, Zalogoyev, S. 1971. Life Sci. Space Res. 9: 11-16. 3. Spizizen, J. 1971. Life Sci. Space Res. 9: 66-68. 4. Mishra, S., Pierson, D. 1992. Encyclopedia of Microbiology.Vol. 4. pp. 53-60. 5. Cioletti, L., Pierson, D., Mishra, S. 1991. SAE Technical Paper Series No.911512. 6. Belay, T. et al. 2002. J. Allergy Clin. Immunol. 110(2): 262-268. 7. Tixador R. et al. 1985. Acta Astronautica. 12: 131-134. 8. Puleo, J.R., Oxborrow, G.S., Fields, N.D., Hall, H.E. 1970. Appl. Microbiol. 20 (3): 384-389.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from objective 1 of this study will allow us to understand how the space flight environment can impact the global gene expression of respiratory pathogens such as Streptococcus pneumoniae and the subsequent impact on its virulence potential. These results will advance our understanding of how changes in gene expression may influence the virulence mechanisms used by S. pneumoniae as it adapts to different environments such as those experienced in space. Results from objective 2 will provide important information on how space environment characteristics such as low-shear and reduced gravitational forces can affect the physiology of disease-associated clinical isolates (e.g. adherent-invasive E. coli; AIEC) which could potentially contribute to their associated disease characteristics. Understanding the genomic responses of S. pneumoniae and the physiological responses of uncharacterized clinical isolates such as AIEC and in the space environment will aid in the development of risk assessments to address potential threats which could contribute to human disease during space flight. Information gained from this study will assist in understanding the threat of opportunistic infections which will be important in the planning and execution of long-term space missions and the habitation of space. Results from this study will contribute to the long-term goals of NASA's human space exploration directive and lead to the development of effective countermeasures crew members could use to detect and treat infections during space flight. The direct earth benefits gained from the objectives of this study include: a.) gaining new knowledge about S. pneumoniae pathogenic mechanisms which could lead to the development of novel vaccines and therapeutics for this pathogen and b.) new information about how uncharacterized, disease-associated clinical isolates such as AIEC respond to mechanically-based forces such as low-shear environments as encountered in vivo such as in the respiratory and gastrointestinal tracts of the body. Such findings could provide valuable insight which could lead to the development of novel vaccines and therapeutics.
	Habitation in space exposes both humans and microbes to microgravity conditions which are characterized by reductions in fluid shear forces. Areas of low-shear stress are also encountered in physiologically relevant regions of the body including the respiratory, gastrointestinal, and urogenital tracts. Conditions of low-shear and modeled microgravity (LSMMG) can be modeled using rotating-wall bioreactors known as high-aspect ratio vessels (HARVs). The LSMMG environment has been shown to impact both bacterial physiology and virulence properties. The effects of LSMMG were investigated on Streptococcus pneumoniae global gene expression by microarray analysis (Objective 1) and on adherent-invasive Escherichia coli (AIEC) physiology and colonization properties by (Objective 2) to better understand how this unique environment can impact different bacterial species influencing their gene expression and physiology. (Objective 1) Previous studies have evaluated the global transcriptional profiles of Gram-negative bacteria; however, no Gram-positive species have been examined. In an effort to identify trends in the global gene expression of S. pneumoniae in response to the LSMMG environment, hierarchical cluster analysis was carried out on microarray results generated from HARV-grown cultures. This analysis revealed a dramatic down-shift in gene expression of S. pneumoniae strain TIGR4 (serotype 4), after growth under LSMMG compared to 1 x g (normal gravity) controls. Next, statistical analysis was carried out to identify genes responsive to the LSMMG environment which were significantly altered. Statistical analyses revealed that the expression of 81 genes was significantly altered. These genes were found to be associated with 7 different functional categories, including many which were uncharacterized. The differential

expression of these various genes indicated that growth of S. pneumoniae under LSMMG affects genes associated with many different physiological functions, including metabolism, signaling, DNA repair/recombination, surface protein expression, and ion transport. These differentially expressed genes were physically located throughout the genome, which suggests that LSMMG conditions impact global transcriptional activity encompassing the entire chromosome. Microarray results were independently confirmed by measuring gene expression changes of seven down-regulated genes randomly selected and analyzed by quantitative real-time RT-PCR. The real-time RT-PCR results confirmed decreased expression of all 7 genes, in concordance with the microarray results.

Among the LSMMG-responsive down-regulated genes were members of common operons and/or clusters involved in competence (DNA uptake), antimicrobial peptide production, and carbohydrate transport. Two down-regulated genes (cglB and cglC) identified are members of the cgl operon, which is one of the late competence operons induced during the onset of competence in the S. pneumoniae life cycle (1-2). The proteins encoded by these genes are involved in construction of the surface-expressed translocasome (DNA uptake apparatus) that functions in DNA uptake and processing during competence (2). The microarray results revealed the down-regulation of three genes (blpL, blpX, and blpY) which are members of the blp (bacteriocin-like peptide) regulon, a quorum-sensing system which functions in the production of antimicrobial peptides and immunity proteins which are co-transcribed for intra- and interspecies competition during colonization. Producer strains of bacteriocins are protected through the synchronous production of immunity proteins which associate with the target surface receptors preventing pore formation (3). The blpL, blpX, and blpY genes are members of operons which encode the protective immunity proteins. Previous reports have shown evidence of cross-talk between the competence and Blp pathways (4). These findings indicate the potential for cross-talk between two independent quorum-sensing systems containing genes responsive to the LSMMG environment and the possibility of convergent regulatory mechanisms. Another group of down-regulated genes included five members of the phosphotransferase systems (PTS), involved in carbohydrate uptake and metabolism. Previous reports have shown that PTS pathway components can serve as a targets for antimicrobial peptides (5). Thus, the down-regulation of genes encoding PTS components may serve as a survival strategy during colonization under LSMMG. Previous studies in the laboratory of Dr. David Niesel have shown that S. pneumoniae exhibits an increased ability to adhere to and invade A549 respiratory epithelia after growth under LSMMG conditions (unpublished data). Our microarray analysis revealed that the genes encoding the various factors associated with these processes were not significantly affected by growth under LSMMG conditions. Thus, no direct correlations were identified which could explain the altered physiological properties observed. While no known virulence genes were found to be induced under LSMMG, the repression of certain virulence genes may play an important functional role in immune evasion, thus contributing to the survival and pathogenesis of the bacteria.

Task Progress:

(Objective 2) While previous studies examining the effects of LSMMG on bacteria have focused on well-characterized strains of both commensal and pathogenic species, there is limited information regarding the effects of LSMMG on disease-associated clinical isolates such as adherent-invasive E. coli strains associated with Crohn's Disease (form of inflammatory bowel disease). In an effort to characterize the effects of LSMMG on the physiology and colonization potential of such isolates, AIEC strain O83:H1 was grown under LSMMG and then analyzed for changes in environmental stress resistance and adherence/invasion capabilities. To determine to potential role of the global stress regulator RpoS under LSMMG, an rpoS mutant (CAA001) was generated and also examined. Analysis of wild-type AIEC strain O83:H1 and CAA001, after growth under LSMMG, revealed alterations in environmental stress resistance including increased thermal stress resistance and rpoS-dependent oxidative and acid stress resistance. While both strains exhibited enhanced adherence after growth under LSMMG conditions, CAA001 displayed a hyper-adherent phenotype which was significantly greater than the wild-type strain. In order to identify putative RpoS-regulated proteins involved in this phenotype, transposon (TnphoA) mutagenesis was carried out on CAA001 used generate mutants in which the hyper-adherent phenotype was abolished under LSMMG. In one of the generated mutants (CAA003) the transposon insertion was mapped within the tnaB gene, which encodes tryptophan permease (imports tryptophan). Complementation of this mutant with an intact tnaB gene restored adherence capabilities. To further investigate the molecular mechanism by which the tnaB gene and its associated (tryptophanase) operon might be functioning, LSMMG-wild-type and CAA003 cultures were supplemented with indole and adherence was evaluated. Indole is a byproduct of tryptophan metabolism that is exported outside the cell and has been suggested to function as an extracellular signaling molecule involved in adherence (6-7). Supplementation with exogenous indole restored wild-type adherence function in CAA003 cultures grown under LSMMG thus suggesting a potential role and mechanism for tnaB and the tryptophanase operon in AIEC adherence under LSMMG.

The research findings from the objectives of this study show how conditions of the space environment, such as microgravity and low-shear stress, can alter the genomic profile of Gram-positive bacteria (e.g. S. pneumoniae) as well as the physiology and colonization properties of disease-associated clinical isolates (e.g. AIEC). The genomic responses of S. pneumoniae to the LSMMG environment could suggest that the lack of virulence gene expression might contribute to immunoevasive strategies elicited by the bacteria to avoid detection by the host immune system. Alternatively, LSMMG-mediated responses could be promoting the induction of novel genes and mechanisms associated with virulence function and / or the fine-tuning of characterized genes and mechanisms in a different way to promote virulence-associated functions. Physiological analyses of AIEC strain O83:H1 after growth under LSMMG revealed both alterations in environmental stress resistance and colonization potential. The absence of rpoS expression under LSMMG conditions conferred a hyper-adherent phenotype which was found to involve participation of the tryptophan permease (tnaB) through a possible signaling mechanism involving the tryptophan metabolite indole. These findings extend our understanding of how the spaceflight environment (e.g. LSMMG) and comparable mechanical forces experienced in vivo can affect the functions of Gram-positive and Gram-negative species.

References:

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